THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If tangents PA and PB from an external point P to a circle with centre O are inclined to each other at 80°, then $\angle POA$ is: 1

(a) 40° **(b)** 50° **(c)** 80° **(d)** 100°

2. A point P is 34 cm from the centre O of a circle of radius 16 cm. The length of the tangent drawn from P to the circle is: 1

(a) 30 cm **(b)** 18 cm **(c)** 50 cm **(d)** 900 cm

3. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 9$ cm, $BC = 11$ cm and $CD = 7$ cm. The length of AD is: 1

(a) 13 cm **(b)** 5 cm **(c)** 27 cm **(d)** 9 cm

4. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 60^\circ$. The $\angle OAB$ is: 1

(a) 29° **(b)** 31° **(c)** 60° **(d)** 30°

5. A tangent to a circle of radius 9 cm has length 40 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 41 cm **(b)** 82 cm **(c)** 49 cm **(d)** 31 cm

6. In $\triangle ABC$, right-angled at B, $AB = 6$ cm and $BC = 8$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 3 cm **(b)** 5 cm **(c)** 4 cm **(d)** 2 cm

7. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 5)$ cm and $QB = (2x + 9)$ cm. The length of QA is: 1

(a) 4 cm **(b)** 19 cm **(c)** 21 cm **(d)** 17 cm

8. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. The length of PQ is: 1

(a) 12 cm **(b)** 8.5 cm **(c)** $\sqrt{119}$ cm **(d)** 13 cm

9. A point P is 35 cm from the centre O of a circle of radius 21 cm. The length of the tangent drawn from P to the circle is: 1

(a) 56 cm **(b)** 14 cm **(c)** 28 cm **(d)** 784 cm

10. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 12$ cm, $BC = 10$ cm and $CD = 8$ cm. The length of AD is: 1

(a) 14 cm **(b)** 30 cm **(c)** 10 cm **(d)** 6 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 4)$ cm and $QB = (2x + 9)$ cm. Find the length of QA. 2

12. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 40°$. Find the $\angle OAB$. 2

13. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 13$ cm, $BC = 9$ cm and $CD = 8$ cm. Find the length of AD. 2

14. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (5x + 1)$ cm and $QB = (3x + 9)$ cm. Find the length of QA. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 12$ cm, $BC = 7$ cm and $CA = 9$ cm. Find the lengths AF, BD and CE (in this order). 3

16. In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

17. Two concentric circles have radii 5 cm and 3 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 3

OR

The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 14$ cm, $BC = 10$ cm and $CA = 12$ cm. Find the lengths AF, BD and CE (in this order). 3

18. In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 32$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

OR

Two concentric circles have radii 13 cm and 5 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. 5

OR

In $\triangle ABC$, $\angle B = 90°$, $AB = 6$ cm and $BC = 8$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

20. Prove that the lengths of the tangents drawn from an external point to a circle are equal. 5

OR

In $\triangle ABC$, $\angle B = 90°$, $AB = 5$ cm and $BC = 12$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

END OF QUESTION PAPER **Total: 40 Marks**